Solutions Tutorial 4

1. Find y' and y'' using implicit differentiation for a function $y(x)$ that is defined by equation $y-\varepsilon\sin y=x$ where $0<\varepsilon<1$ is a fixed constant.

Solution. Implicitly differentiate $y-\varepsilon\sin y=x$ with respect to x:

$$y'-(\varepsilon\cos y)\times y'=1,\text{ or }y'=\frac{1}{1-\varepsilon\cos y}.$$

Implicitly differentiate with respect to x for the second time: $y''=-\frac{\varepsilon y'\sin y}{(1-\varepsilon\cos y)^2}$.

Substitute $y'=\frac{1}{1-\varepsilon\cos y}$ into the above equation to obtain: $y''=-\frac{\varepsilon\sin y}{(1-\varepsilon\cos y)^3}$.

2. Suppose that z is defined implicitly by $x^3+y^3+z^3+3xyz=5$. Find z_x and z_y using implicit differentiation.

Solution. $3x^2+3z^2z_x+3(yz+xyz_x)=0\Rightarrow z_x=-\frac{x^2+yz}{xy+z^2}$ and $z_y=-\frac{y^2+xz}{xy+z^2}$ (symmetry).

Implicit function theorem (special case)

Let $f(x,y,z):\mathbb{R}^3\to\mathbb{R}$ be a function of class C^1 where x and y are independent variables and z is a function in x and y. If there exists a solution point $(a,b,c)\in\mathbb{R}^3$ (i.e., $f(a,b,c)=0$) such that $f_z(a,b,c)\neq 0$. Then there exists an explicit function $z=\phi(x,y):\mathbb{R}^2\to\mathbb{R}$ near to (a,b,c) such that $\phi(a,b)=c$.

3. Let $G:\mathbb{R}\times\mathbb{R}^+\times\mathbb{R}^+\to\mathbb{R}$ be a function defined as $G(x,y,z)=xy+xz\ln(yz)=1$. Show that (1,1,1) is a solution point. Show that there exists a solution function $z=z(x,y)$ near the point (1,1,1) with $z(1,1)=1$. Find a formula for $z_x(x,y)$ and evaluate $z_x(1,1)$.

Solution. Let $f(x,y,z)=xy+xz\ln(yz)-1\Rightarrow f(1,1,1)=1+0-1=0$. So (1,1,1) is a solution point. Now, partially differentiate f with respect to x, y and z:

$$f_x=y+z\ln(yz),\qquad f_y=x+\frac{xz}{y},\quad f_z=x\ln(yz)+xz\times\frac{y}{yz}=x\ln(yz)+x$$

$$\text{and }f_z(1,1,1)=1(0)+1=1\neq 0.$$

Note that f is in C^1 class and we see that the above fulfil the implicit function theorem. So, we conclude that a solution function $z=z(x,y)$ exists near the point (1,1,1) with $z(1,1)=1$. We implicitly differentiate the equation $xy+xz\ln(yz)=1$ with respect to x on both sides:

$$y+z\ln(yz)+xz_x\ln(yz)+xz\times\frac{y}{yz}\times z_x=0\Rightarrow z_x=-\frac{y+z\ln(yz)}{x\ln(yz)+x}\Rightarrow z_x(1,1)=-1.$$

4. Suppose that u, v are defined implicitly (as functions of x and y) by the pair of equations $u^2 - v = 3x + y$ and $u - 2v^2 = x - 2y$. Find u_x and v_x in terms of u and v.

Solution. $\begin{cases} u^2 - v = 3x + y \\ u - 2v^2 = x - 2y \end{cases} \overset{\frac{\partial}{\partial x}}{\Rightarrow} \begin{cases} 2uu_x - v_x = 3 \quad (1) \\ u_x - 4vv_x = 1 \quad (2) \end{cases} \overset{(1)\times 4v-(2):}{\overset{(2)\times 2u-(1):}{\Rightarrow}} \begin{cases} u_x = \frac{1-12v}{1-8uv} \\ v_x = \frac{2u-3}{1-8uv} \end{cases}$.

5. Consider the equation $(x^4 + y^2 + z^2)^{\frac{1}{2}} - \cos x - \cos y = 0$. Explain why this equation implicitly determines z as a continuously differentiable function of x and y near the point $(0,0,2)$. Hence, find $z_x(0,0)$ and $z_y(0,0)$.

Solution. Let $f(x,y,z) = (x^4 + y^2 + z^2)^{\frac{1}{2}} - \cos x - \cos y$. Then

$$f_x = \tfrac{1}{2}(x^4 + y^2 + z^2)^{-\frac{1}{2}}(4x^3) + \sin x, \quad f_y = \tfrac{1}{2}(x^4 + y^2 + z^2)^{-\frac{1}{2}}(2y) + \sin y,$$
$$f_z = \tfrac{1}{2}(x^4 + y^2 + z^2)^{-\frac{1}{2}}(2z) \Rightarrow f_z(0,0,2) = \tfrac{2(2)}{2(2)} = 1 \neq 0$$

Note that $f(0,0,2) = 0$ and f is in C^1 class. So $(0,0,2)$ is a solution point. Hence, the implicit function theorem guarantees the existence of a solution function $z = z(x,y)$ near the point $(0,0)$ with $z(0,0) = 2$. Note that the equation is symmetry for x and y. Implicitly differentiate $(x^4 + y^2 + z^2)^{\frac{1}{2}} - \cos x - \cos y = 0$ with respect to x:

$$\tfrac{4x^3 + 2zz_x}{2}(x^4 + y^2 + z^2)^{-\frac{1}{2}} + \sin x = 0 \Rightarrow \tfrac{0 + 2(2)z_x(0,0)}{2} \times \tfrac{1}{2} = 0 \Rightarrow z_x(0,0) = 0.$$

So $z_y(0,0) = 0$ also.

Alternative: If we have $f(x,y,z(x,y)) = 0$, then $z_x = -\frac{f_x}{f_z}$ and $z_y = -\frac{f_y}{f_z}$.

Proof. $f(x,y,z(x,y)) = 0$. Implicitly differentiate with respect to x:

$$\begin{cases} f_x + f_z z_x = 0 \\ f_y + f_z z_y = 0 \end{cases} \Rightarrow \begin{cases} z_x = -\frac{f_x}{f_z} \\ z_y = -\frac{f_y}{f_z} \end{cases}.$$

So, we can find $z_x(0,0) = -\frac{f_x(0,0)}{f_z(0,0)} = -\left.\frac{2x^3\left(x^4+y^2+z^2\right)^{-\frac{1}{2}} + \sin x}{z\left(x^4+y^2+z^2\right)^{-\frac{1}{2}}}\right|_{(0,0,2)} = 0$ and $z_y(0,0) = 0$ also.

6. What does the implicit function theorem say about the following equation

$$x^2 + y^3 + z^4 + z^2 \sin x \cos y = 2 \text{ near } (1,1,0)\,?$$

Solution. We consider the following cases:

Case 1. $x = \phi(y,z)$ (y and z independent and x dependent)

Case 2. $y = \varphi(x,z)$ (x and z independent and y dependent)

Case 3. $z = \psi(x,y)$ (x and y independent and z dependent)

Case 4. $x = f(z), y = g(z)$ (z is the independent variable)

Case 5. $x = s(y), z = t(y)$ (y is the independent variable)

Case 6. $y = u(x), z = v(x)$ (x is the independent variable)

Define $f(x,y,z) = x^2 + y^3 + z^4 + z^2 \sin x \cos y - 2$.

Note that $f(1,1,0) = 0$, and

$$\begin{aligned} f_x &= 2x + z^2 \cos x \cos y && \Rightarrow f_x(1,1,0) = 2 \neq 0 \\ f_y &= 3y^2 - z^2 \sin x \sin y && \Rightarrow f_y(1,1,0) = 3 \neq 0 \\ f_z &= 4z^3 + 2z^2 \sin x \cos y && \Rightarrow f_z(1,1,0) = 0 \end{aligned}$$

Therefore, the implicit function theorem concludes that there exists a solution function $x = \phi(y,z)$, $y = \varphi(x,z)$ ((near $(1,1,0)$ for both cases); but it does not guarantee the existence of a solution function $z = \psi(x,y)$.

Remark: Case 4 – 6 cannot be applied because the total derivatives formed is not a square matrix and thus, we cannot determine whether it is invertible or not, which mean in those cases, the implicit function theorem cannot be applied.

Extra Exercises

7. Let $f(x,y,z):\mathbb{R}^3\to\mathbb{R}$ be a function such that y and z are defined implicitly, as functions of x, by $f(x,y,z)=0$ and $x^2+y^2+z^2=1$. Show that $\frac{dy}{dx}=-\frac{zf_x-xf_z}{zf_y-yf_z}$.

Solution. $x^2+y^2+z^2=1\Rightarrow 2x+2y\frac{dy}{dx}+2z\frac{dz}{dx}=0\Rightarrow \frac{dz}{dx}=-\frac{1}{z}\left(x+y\frac{dy}{dx}\right)\quad(1)$

$f(x,y,z)=0\Rightarrow \frac{df}{dx}=f_x+f_y\times\frac{dy}{dx}+f_z\times\frac{dz}{dx}=0\underset{\text{sub (1)}}{\Rightarrow} f_x+f_y\times\frac{dy}{dx}-f_z\times\frac{1}{z}\left(x+y\frac{dy}{dx}\right)=0$

So $\frac{dy}{dx}=-\frac{zf_x-xf_z}{zf_y-yf_z}$ and $\frac{dz}{dx}=-\frac{yf_x-xf_y}{yf_z-zf_y}$ (symmetry, swap y and z.)

8. Suppose that u, v are defined implicitly as functions of x and y by the pair of equations $u^2+v=x+y$ and $u+v^2=x-y$. Show that $u_xu_y-v_xv_y=\frac{2(2u^2+2v^2-1)}{(4uv-1)^2}$.

Solution.
$$\begin{cases}u^2+v=x+y\\ u+v^2=x-y\end{cases}\underset{\frac{\partial}{\partial x}}{\Rightarrow}\begin{cases}2uu_x+v_x=1\ (1)\\ u_x+2vv_x=1\ (2)\end{cases}\underset{(2)\times 2u-(1):}{\overset{(1)\times 2v-(2):}{\Rightarrow}}\begin{cases}u_x=\frac{2v-1}{4uv-1}\\ v_x=\frac{2u-1}{4uv-1}\end{cases}$$
$$\begin{cases}u^2+v=x+y\\ u+v^2=x-y\end{cases}\underset{\frac{\partial}{\partial y}}{\Rightarrow}\begin{cases}2uu_y+v_y=1\ (3)\\ u_y+2vv_y=-1\ (4)\end{cases}\underset{(4)\times 2u-(3):}{\overset{(3)\times 2v-(4):}{\Rightarrow}}\begin{cases}u_y=\frac{2v+1}{4uv-1}\\ v_y=-\frac{2u+1}{4uv-1}\end{cases}$$
$$\therefore u_xu_y-v_xv_y=\frac{1}{(4uv-1)^2}\Big((2v-1)(2v+1)+(2u-1)(2u+1)\Big)=\frac{2(2u^2+2v^2-1)}{(4uv-1)^2}$$

9. Consider the pair of equations $\begin{cases}u^2+vw+w^3-3=0\\ 2u^3-uvw+w^2-2=0\end{cases}$. Suppose we are interested in solutions near the obvious solution $(u,\,v,\,w)=(1,\,1,\,1)$. Suppose u and v are function of w. Use the implicit function theorem to determine if there exists a solution function for $(u,\,v)$ as a function of w.

Solution. Let $f(w,u,v)=\begin{pmatrix}u^2+vw+w^3-3\\ 2u^3-uvw+w^2-2\end{pmatrix}$ where w is independent and u, v are dependent. We may write $y=(u,v)=(u(w),v(w))$. Let $f_1=u^2+vw+w^3-3=0$, $f_2=2u^3-uvw+w^2-2=0$. We see that f_1,f_2 are continuous and all the first partial derivatives exist and are all continuous:

$$\frac{\partial f_1}{\partial u}=2u,\qquad \frac{\partial f_1}{\partial v}=w,\qquad \frac{\partial f_1}{\partial w}=v+3w^2$$
$$\frac{\partial f_2}{\partial u}=6u^2-vw,\qquad \frac{\partial f_2}{\partial v}=-uw,\qquad \frac{\partial f_2}{\partial w}=-uv+2w$$

We see that $f(1,1,1)=\begin{pmatrix}0\\0\end{pmatrix}$ and $\det\begin{pmatrix}2u & w\\ 6u^2-vw & -uw\end{pmatrix}\Big|_{u=v=w=1}=\begin{vmatrix}2 & 1\\ 5 & -1\end{vmatrix}=-7\neq 0$. So, we conclude that there exists solution $y:\mathbb{R}^2\to\mathbb{R}$ near to $w=1$ such that $y(1,\,1)=1$.

10. What does the implicit function theorem say about the following equation

$$\begin{cases} xy + yz^2 = 1 \\ x^2 + y^3 = 2 \end{cases} \text{ near } (1,1,0)?$$

Solution. Define $f(x,y,z) = \begin{pmatrix} f_1(x,y,z) \\ f_2(x,y,z) \end{pmatrix} = \begin{pmatrix} xy + yz^2 - 1 \\ x^2 + y^3 - 2 \end{pmatrix}$ at $(1,1,0)$.

$$\begin{aligned} \tfrac{\partial f_1}{\partial x} &= y, & \tfrac{\partial f_1}{\partial y} &= x + z^2, & \tfrac{\partial f_1}{\partial z} &= 2yz \\ \tfrac{\partial f_2}{\partial x} &= 2x, & \tfrac{\partial f_2}{\partial y} &= 3y^2, & \tfrac{\partial f_2}{\partial z} &= 0 \end{aligned}$$

We consider several cases (same as Q6). The total derivatives are (for Case 4 – 6)

$$D_{(x,y)}f = \begin{pmatrix} y & x+z^2 \\ 2x & 3y^2 \end{pmatrix}, \quad D_{(x,z)}f = \begin{pmatrix} y & 2yz \\ 2x & 0 \end{pmatrix}, \quad D_{(y,z)}f = \begin{pmatrix} x+z^2 & 2yz \\ 3y^2 & 0 \end{pmatrix}$$

$$D_{(x,y)}f = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}, \qquad D_{(x,z)}f = \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix}, \qquad D_{(y,z)}f = \begin{pmatrix} 1 & 0 \\ 3 & 0 \end{pmatrix}$$

$$\det D_{(x,y)}f = 1 \neq 0 \qquad \det D_{(x,z)}f = 0 \qquad \det D_{(y,z)}f = 0$$

The first matrix is invertible, but the other two are not, so the implicit function theorem says we can solve the pair of equations for (x, y) in terms of z, but does not say if we can solve for (x, z) in terms of y, or (y, z) in terms of x.

Remark: Case 1 – 3 cannot be applied because the total derivatives formed is not a square matrix and so we cannot determine whether it is invertible or not, which mean in those cases, the implicit function theorem cannot be applied.

Past years test/final exam questions

11. (a) Let $f:\mathbb{R}\to\mathbb{R}$ be twice differentiable and $h(x,y)=f(x)f(y^2)+f(xy)$. Find the h_{xy} in terms of f' and f''. Then, evaluate $h_{xy}(1,1)$ if given that $f'(1)=f''(1)=1$.

(b) Let $g:\mathbb{R}^2\to\mathbb{R}$ be a function defined as $g(u,v)=u^2+v^2$ with $u:\mathbb{R}^2\to\mathbb{R}$ and $v:\mathbb{R}\times\mathbb{R}^+\to\mathbb{R}$ defined as $u(x,y)=xe^y$, $v(x,y)=x\ln y$. Write $\frac{\partial g}{\partial x}$ and $\frac{\partial g}{\partial y}$ in terms of x and y.

Solution. (a) $h_x=f'(x)f(y^2)+yf'(xy)\Rightarrow h_{xy}=2yf'(x)f'(y^2)+f'(xy)+xyf''(xy)$

$h_{xy}(1,1)=2+1+1=4$

(b) $\frac{\partial g}{\partial x}=g_u u_x+g_v v_x=2ue^y+2v\ln y=2xe^{2y}+2x(\ln y)^2$

$\frac{\partial g}{\partial y}=g_u u_y+g_v v_y=2uxe^y+2v\frac{x}{y}=2x^2e^{2y}+2\left(\frac{x^2}{y}\right)\ln y$

12. Let $f:\mathbb{R}^3\to\mathbb{R}$ be a function and assume that there exist functions $y,z:\mathbb{R}\to\mathbb{R}$ such that $f(x,y(x),z(x))=0$.

(a) Suppose $f_y(x,y(x),z(x))\neq 0$. Prove that $\frac{dy}{dx}=-\frac{f_x}{f_y}-\frac{f_z\frac{dz}{dx}}{f_y}$.

(b) Let $f:\mathbb{R}\times\mathbb{R}^+\times\mathbb{R}^+\to\mathbb{R}$ defined as $f(x,y,z)=xy+z\ln(yz)$. Show that $(0,1,1)$ is a solution point, and that $f_y(0,1,1)\neq 0$.

(c) Assume that $y(x)$ and $z(x)$ exist in a neighbourhood of the point $(0,1,1)$, and that $\left.\frac{dy}{dx}\right|_{(0,1,1)}=\left.\frac{dz}{dx}\right|_{(0,1,1)}=C$. Use the result of part (a) to show that $C=-\frac{1}{2}$. Explain why the use of part (a) is justified.

Solution. (a) Given $f_y(x,y(x),z(x))\neq 0$. From $f(x,y(x),z(x))=0$, we differentiate with respect to x:

$$f_x+f_y\frac{dy}{dx}+f_z\frac{dz}{dx}=0 \Rightarrow \frac{dy}{dx}=-\frac{f_x+f_z\frac{dz}{dx}}{f_y}$$

$f(0,1,1)=0(1)+1\ln 1=0$ $\quad\boxed{f_y=x+\frac{z}{y}}\Rightarrow f_y(0,1,1)=1\neq 0$

(c) $\boxed{f_x=y}\Rightarrow f_x(0,1,1)=1$ and $\boxed{f_z=1+\ln(yz)}\Rightarrow f_z(0,1,1)=1$.

At $(0,1,1)$: $\frac{dy}{dx}(0,1,1)=-\frac{f_x(0,1,1)}{f_y(0,1,1)}-\frac{f_z(0,1,1)\times\frac{dz}{dx}(0,1,1)}{f_y(0,1,1)}\Rightarrow C=-\frac{1+C}{1}\Rightarrow C=-\frac{1}{2}$.

Because $f(0,1,1)=0$ and $f_y(0,1,1)\neq 0$.

13. Let $z(x,y):\mathbb{R}^2\to\mathbb{R}$ be a function satisfying $x^2+(y-z)^2=1$ and $z(x,y)\neq y$, where x and y are independent variables.

(a) Show that $z_{xy}(x,y)=z_{yx}(x,y)$ (without using the mixed derivative theorem).

(b) Hence, evaluate $z_{xy}(\pi,-1)$.

Solution. $x^2+(y-z)^2=1$

(a) $\underset{\frac{\partial}{\partial y}}{\Rightarrow} 2(y-z)(1-z_y)=0\Rightarrow z_y=1\because z(x,y)\neq y\Rightarrow z_{yx}=0^{\#}$

$\underset{\frac{\partial}{\partial x}}{\Rightarrow} 2x-2(y-z)z_x=0^{*}\Rightarrow z_x=\frac{x}{y-z}, z\neq y$

From*, we partially differentiate with respect to y:

$(1-z_y)z_x+(y-z)z_{xy}=0\Rightarrow z_{xy}=\frac{(1-z_y)z_x}{z-y}=0\because z_y=1, z\neq y$

$\therefore z_{xy}=0=z_{yx}$ from $^{\#}$

(b) $z_{xy}(\pi,-1)=0$ in fact $z_{xy}(x,y)=0$ for any x and y.

14. (a) State the Implicit Function Theorem.

(b) Consider the equation $x\sin z+y\cos z=1$. By using the Implicit Function Theorem, determine whether it is possible to define an explicit function $z(x,y)$ in the neighbourhood of $(x,y)=(1,1)$ with $z(1,1)=0$. Justify your answer.

Solution. (a) Let $f(x,y(x)):\mathbb{R}^{m+n}\to\mathbb{R}^n$ be a function such that $f\in C^1$, where $x\in\mathbb{R}^m$ are independent variables and $y(x)\in\mathbb{R}^n$ are dependent variables. If there exists $(a,b)\in\mathbb{R}^{m+n}$ such that $f(a,b)=0$ and D_yf is invertible. Then there exists an explicit function $y(x):\mathbb{R}^m\to\mathbb{R}^n$ close to a such that $y(a)=b$ and $f(x,y(x))=0$.

(b) $f(x,y,z)=x\sin z+y\cos z-1=0\Rightarrow f_x=\sin z,\ f_y=\cos z,\ f_z=x\cos z-y\sin z$

(i) All f_x,f_y,f_z exist and continuous at $(1,1,0)$.

(ii) There exists $(1,1,0)\in\mathbb{R}^3$ such that $f(1,1,0)=0$; and

(iii) $f_z(1,1,0)=1\neq 0$.

This proves the existence of explicit function $z(x,y)$ in the neighbourhood of $(1,1)$ with $z(1,1)=0$.

15. Let $f(u,v)$ be a differentiable function of two variables, and let z be a differentiable function of x and y defined implicitly by $f\left(\frac{z}{x},\frac{z}{y}\right)=0$, $x,y\neq 0$. Given that $x^{-1}f_u+y^{-1}f_v\neq 0$, prove that $xz_x+yz_y=z$.

Proof. Let $u=x^{-1}z, v=y^{-1}z$. Given that $x,y\neq 0$ and $x^{-1}f_u+y^{-1}f_v\neq 0$.

$$\begin{cases} u_x=-x^{-2}z+x^{-1}z_x \\ u_y=x^{-1}z_y \end{cases} \qquad \begin{cases} v_x=y^{-1}z_x \\ v_y=-y^{-2}z+y^{-1}z_y \end{cases} \text{ or by symmetry, } x\leftrightarrow y,\ u\leftrightarrow v.$$

$$0=f_u\cdot u_x+f_v\cdot v_x=f_u\cdot\left(-x^{-2}z+x^{-1}z_x\right)+f_v\cdot y^{-1}z_x$$
$$\Rightarrow x^{-1}zf_u=x\left(x^{-1}f_u+y^{-1}f_v\right)z_x \Rightarrow xz_x=\frac{x^{-1}zf_u}{x^{-1}f_u+y^{-1}f_v}$$

$$0=f_u\cdot u_y+f_v\cdot v_y=f_u\cdot x^{-1}z_y+f_v\cdot\left(-y^{-2}z+y^{-1}z_y\right)$$
$$\Rightarrow y^{-1}zf_v=y\left(x^{-1}f_u+y^{-1}f_v\right)z_y \Rightarrow yz_y=\frac{y^{-1}zf_v}{x^{-1}f_u+y^{-1}f_v} \quad \text{(or by symmetry to obtain it)}$$

Hence, $xz_x+yz_y=\frac{x^{-1}f_u+y^{-1}f_v}{x^{-1}f_u+y^{-1}f_v}z=z$.

16. Let $f:\mathbb{R}^2\to\mathbb{R}$ be a function of u and v, where $u,v:\mathbb{R}^2\to\mathbb{R}$ are the functions of (x,y) given by $\left(u,v\right)=\left(xy,\frac{x}{y}\right)$. Let $h(x,y)=f(u(x,y),v(x,y))$.

(a) Find h_x and h_y .

(b) Show that $xh_x+yh_y=2uf_u$ and $xh_x-yh_y=2vf_v$.

Solution. (a) First note that $y\neq 0$. Then

$$h_x=f_uu_x+f_vv_x=f_u\times y+f_v\times\tfrac{1}{y}=yf_u+y^{-1}f_v$$
$$h_y=f_uu_y+f_vv_y=f_u\times x+f_v\times\left(-\tfrac{x}{y^2}\right)=xf_u-xy^{-2}f_v$$

(b) Hence, $xh_x+yh_y=xyf_u+xy^{-1}f_v+xyf_u-xy^{-1}f_v=2xyf_u=2uf_u$, and

$xh_x-yh_y=xyf_u+xy^{-1}f_v-xyf_u+xy^{-1}f_v=2xy^{-1}f_v=2vf_v$.